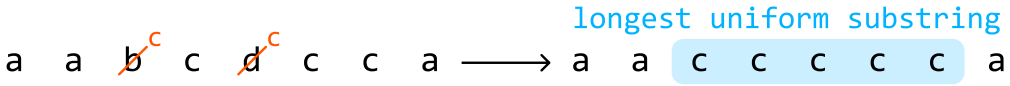
Longest Uniform Substring After Replacements

A **uniform** substring is one in which all characters are identical. Given a string, determine the `length` of the longest uniform substring that can be formed by **replacing up to** `k` **characters**.

Example:




```
Input: s = 'aabcdcca', k = 2
Output: 5
```

Explanation: if we can only replace 2 characters, the longest uniform substring we can achieve is "ccccc", obtained by replacing 'b' and 'd' with 'c'.

Intuition

Determining if a substring is uniform

Before we try finding the longest uniform substring, let's first determine the most efficient way to make a string uniform with the fewest character replacements. Consider the example below:

a b a a b a c

Which characters should we replace to ensure the minimum number of replacements are performed to make the string uniform? There are three main choices: make the string all 'a's, or all 'b's, or all 'c's. The most efficient choice, requiring the fewest replacements, is to make all characters 'a', which involves just three replacements:

a ~~b~~ a a ~~b~~ a ~~c~~ ⟶ a a a a a a a

The key observation is that the minimum number of replacements needed to achieve uniformity is obtained by **replacing all characters except the most frequent one**.

This suggests that if we know the highest frequency of a character in a substring, we can determine if our value of k is sufficient to make that substring uniform. The number of characters that need to be replaced (num_chars_to_replace) can be found by subtracting this highest frequency from the total number of characters in the substring:

a ~~b~~ a a ~~b~~ a ~~c~~

```
num_chars_to_replace = len(substring) - highest_freq
                     = 7 - 4
                     = 3
```

Once we've calculated num_chars_to_replace for a given substring, we can assess if the substring can be made uniform:

- If `num_chars_to_replace` ≤ k, the substring can be made uniform.
- If `num_chars_to_replace` > k, the substring cannot be made uniform.

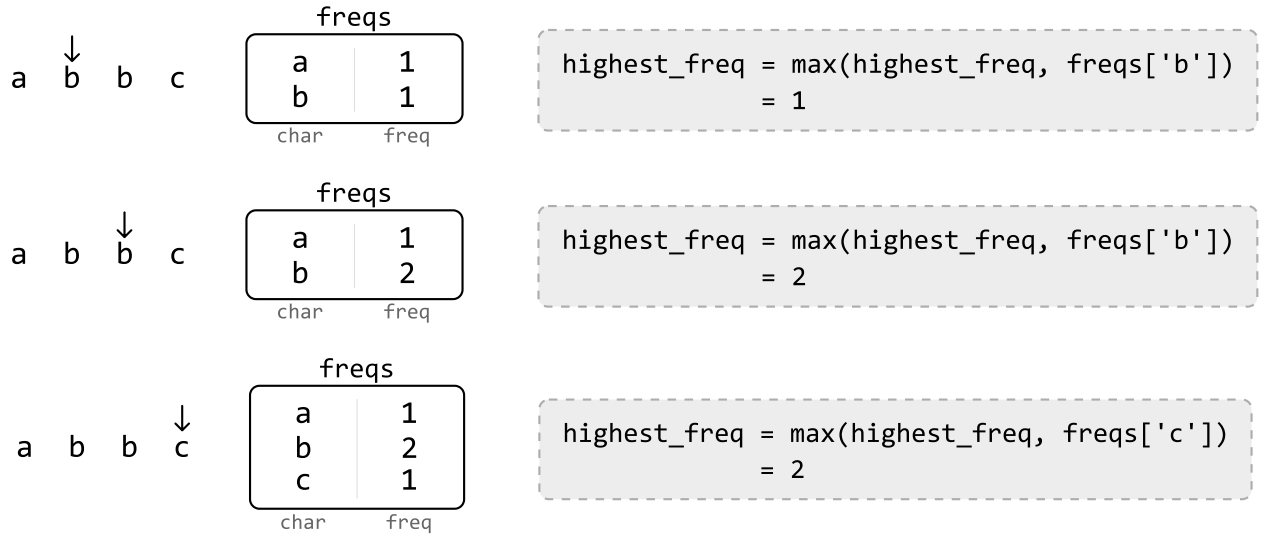
To calculate num_chars_to_replace, we need to know the value of `highest_freq`. This requires tracking the frequency of each character, which can be efficiently managed using a **hash map** (`freqs`). This hash map allows us to update `highest_freq` whenever we encounter a character with a higher frequency. Below is an illustration of how `freqs` is updated:

↓
a b b c

freqs

char	freq
a	1

```
highest_freq = max(highest_freq, freqs['a'])
             = 1
```



Now that we have the tools to determine if a substring can be made uniform, the next step is to figure out how to identify the longest uniform substring. Let's explore a technique that lets us do this.

Dynamic sliding window

We know sliding windows can be useful for solving problems involving substrings. This problem requires that we find the longest substring that satisfies a specific condition:

num_chars_to_replace <= k

So, a dynamic sliding window might be appropriate, as discussed in the chapter introduction.

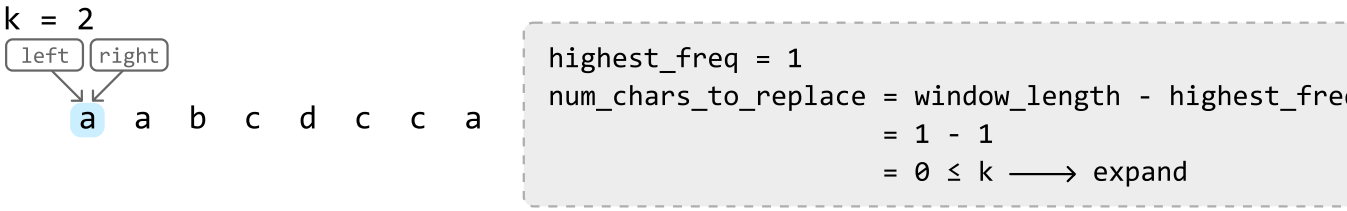
We can use the above condition to determine how to expand or shrink the window:

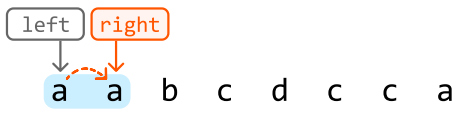
- If the condition is met (i.e., the window is valid), we **expand** the window to find a longer window that still meets this condition.
- If the condition is violated (i.e., the window is invalid), we **shrink** the window until it meets the condition again.

Let's see how this works over the example below:

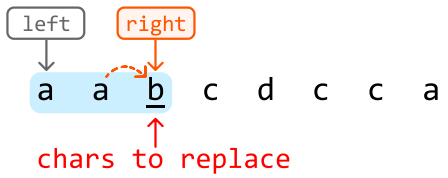
`a a b c d c c a, k = 2`

Start by defining the left and right boundaries of the window at index 0. Continue expanding the window for as long as it satisfies our condition (`nums_char_to_replace <= k`):

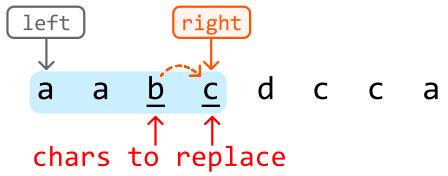





```
highest_freq = 2
num_chars_to_replace = window_length - highest_freq
                     = 2 - 2
                     = 0 ≤ k ⟶ expand
```

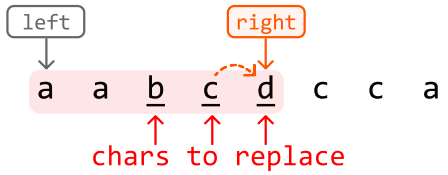



```
highest_freq = 2
num_chars_to_replace = window_length - highest_freq
                     = 3 - 2
                     = 1 ≤ k ⟶ expand
```

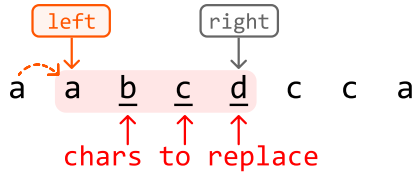



```
highest_freq = 2
num_chars_to_replace = window_length - highest_freq
                     = 4 - 2
                     = 2 ≤ k ⟶ expand
```

Once the window expands to the fifth character ('d'), it will contain 3 characters that must be replaced to make the window uniform. Since we can only replace up to k = 2 characters, the window is invalid. So, we shrink the window:




```
highest_freq = 2
num_chars_to_replace = window_length - highest_freq
                     = 5 - 2
                     = 3 > k ⟶ shrink
```




```
highest_freq = 2 ✕
```

Notice that after shrinking the window, the value of `highest_freq` is still 2, which is no longer correct. Recall that our current method for updating `highest_freq` only increases it when encountering a character with a higher frequency, meaning `highest_freq` can only remain the same or increase, but it can never decrease.

One way to work around this is to develop a new method for updating `highest_freq` that accurately decreases it when the highest frequency in a window decreases. However, our goal is to find the longest substring that meets the condition, so shrinking the window might not even be necessary. The

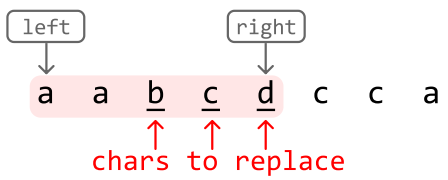
crucial point here is that **when we find a valid window of a certain length, no shorter window will provide a longer uniform substring**.

> This means we can just slide the window instead of shrinking it whenever we encounter an invalid window, effectively maintaining the length of the current window.

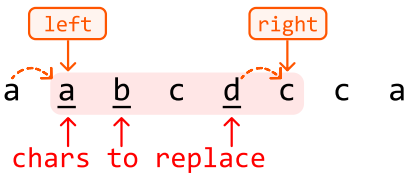
With this observation, we should correct our previous logic:

- If the window satisfies the condition: expand.
- **If the window doesn't satisfy the condition: slide**.

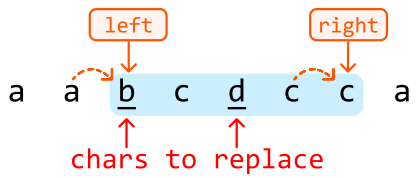
Let's correct the action taken in the first invalid window above by sliding instead of shrinking. Then, we can continue processing the rest of the string.



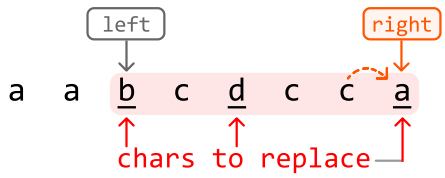

```
highest_freq = 2
num_chars_to_replace = window_length - highest_freq
                     = 5 - 2
                     = 3 > k ⟶ slide
```




```
highest_freq = 2
num_chars_to_replace = window_length - highest_freq
                     = 5 - 2
                     = 3 > k ⟶ slide
```




```
highest_freq = 3
num_chars_to_replace = window_length - highest_freq
                     = 5 - 3
                     = 2 ≤ k ⟶ expand
```




```
highest_freq = 3
num_chars_to_replace = window_length - highest_freq
                     = 6 - 3
                     = 3 < k ⟶ slide
```

The above window is the final window because we cannot expand or slide it any further. The longest valid window encountered during this process is a window of length 5.

Try it yourself

Write your solution to the problem before checking the reference implementation.

Implementation

Python JavaScript Java

```
import java.util.HashMap;

public class Main {
    public int longest_uniform_substring_after_rep
        HashMap<Character, Integer> freqs = new Ha
        int highestFreq = 0;
        int maxLen = 0;
        int left = 0, right = 0;
        while (right < s.length()) {
            // Update the frequency of the charact
            // frequency for the current window.
            char ch = s.charAt(right);
            freqs.put(ch, freqs.getOrDefault(ch, 0
            highestFreq = Math.max(highestFreq, fr
            // Calculate replacements needed for t
            int numCharsToReplace = (right - left
            // Slide the window if the number of r
            // The right pointer always gets advan
            if (numCharsToReplace > k) {
                char leftChar = s.charAt(left);
                freqs.put(leftChar, freqs.get(left
                left++;
            }
            // Since the length of the current win
            // the length of the current window to
            maxLen = right - left + 1;
            // Expand the window.
            right++;
        }
        return maxLen;
    }
}
```

Complexity Analysis

Time complexity: The time complexity of `longest_uniform_substring_after_replacements` is $O(n)$, where n is the length of the input string. This is because we traverse the string linearly with two pointers.

Space complexity: The space complexity is $O(m)$, where m is the number of unique characters in the string stored in the hash map `freqs`.